



C4-MKII Multi-Purpose Teaching Flume

The Armfield Multi-purpose Teaching Flume has been specifically designed to demonstrate the principles of fluid mechanics when applied to engineering structures in open channel flow.

Note:

- The MKII version is supplied as an accessory to the F1-10 Hydraulics Bench and can be used with any recently supplied F1-10
- Available in 2.5m and 5.0m working section lengths
- Clear acrylic sides to give visibility of the working section
- A jacking system permits the slope of the channel bed to be adjusted between -1% and +3%
- Inlet tank with flow stilling arrangement
- Includes a Venturi, sharp and broad-crested weirs, 2 vernier level gauges, adjustable undershot weir and crump weir
- Wide range of other models available as accessories.
- Optional flow meter
- Comprehensive instruction manual

PRODUCT CODE: C4-MKII

Categories: Agricultural, C Series - Advanced Fluid Mechanics, C4-MKII - Multi-Purpose Teaching Flume, Civil, Educational, Environmental

Tags: Air-Regulated Syphon , Applied Fluid Mechanics , Artificially Roughened Bed 2.5m , Bed-Pressure Tappings , C4-MkII , Calibrated Screwjack , Culvert Fitting , Data Processing , False Floor , Flow Meter , Force-Momentum , Graph Plotting , Hook And Point Gauges , Hydraulic Jump , Hydraulic Structures , Hydraulics Bench , Manning's Formula , Measure Water Level , Multi-Purpose Teaching Flume , Open Channel Flow , Open Channel Flume , Overshot Weir , Pitot-Static Tube , Radial Gate , Siphon , Slope Adjustment , Sloping Apron , Sluice Gate , Super-Critical Flow , Syphon Spillways , Underlying Characteristics Of Waves , Veloc

DESCRIPTION

The Armfield Multi-purpose Teaching Flume has been specifically designed to demonstrate the principles of fluid mechanics when applied to engineering structures in open channel flow.

The C4-MKII is a small open channel flume, available in 2.5m or 5.0m lengths, with clear acrylic sides to the working section for complete visibility of the flow.

The channel is fitted with a PVC inlet tank, and is designed for free discharge into the Hydraulics Bench. The flume is mounted on a rigid framework, and can be tilted by use of a calibrated screwjack, which enables accurate slope adjustment of the channel.

The inlet tank incorporates a stilling arrangement to diffuse the water flow prior to entry into the channel, ensuring smooth uniform flow. The level in the working section of the flume is controlled using an overshoot weir (stop logs) at the discharge end.

Bed-pressure tappings and fixing points for models are provided.

A longitudinal scale positioned at the top of the channel enables depth gauges and pitot static tubes to be accurately positioned along the channel length.

The flume is designed for use with a standard Armfield F1-10 Hydraulics Bench, which provides the pumped water flow, the flow control valve and a volumetric tank for flow measurement.

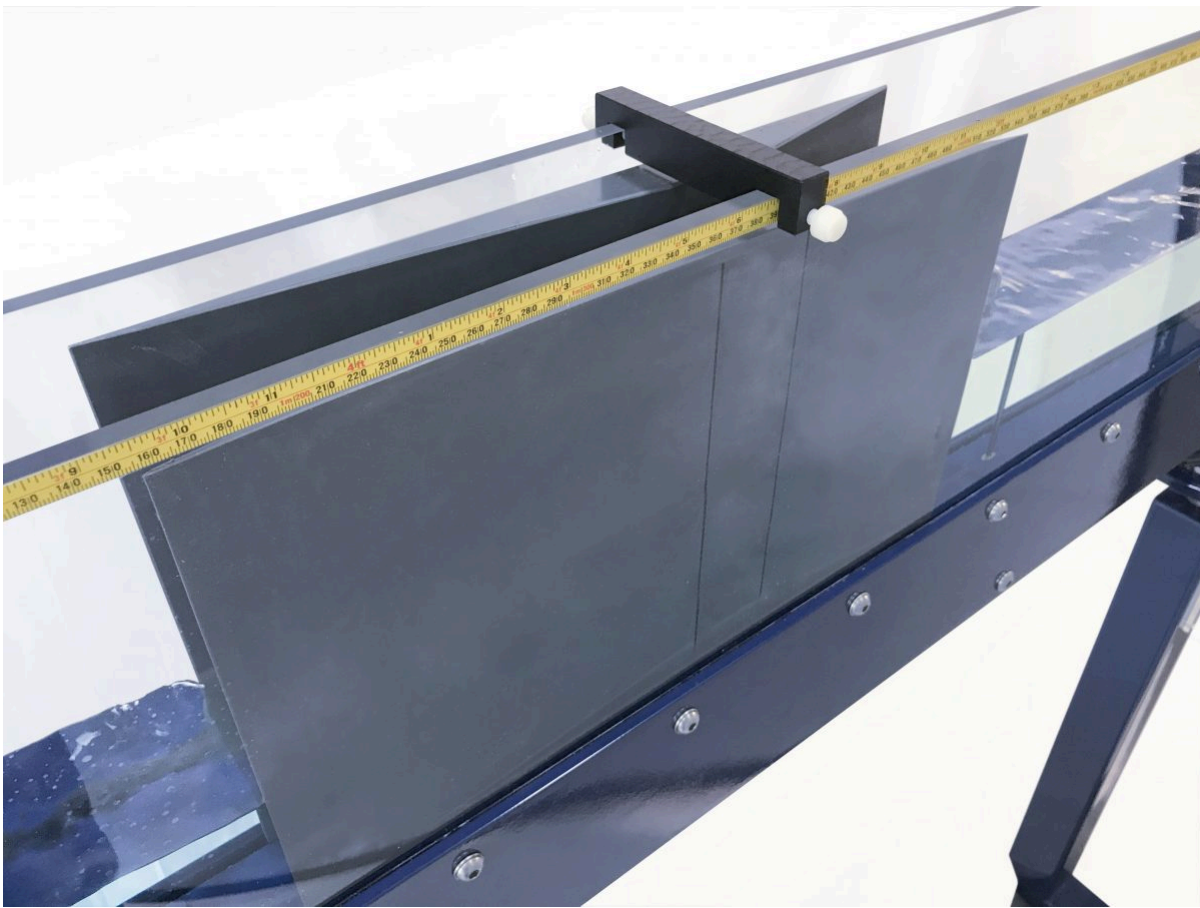
Also available is an optional flow meter, which can be fitted to the C4-MKII to enable direct flow measurements to be taken.

Optional educational software is available (C4-MKII-ABASIC) offering a complete teaching package of coursework. The student manually enters data in the software, which can then be used for calculations, data processing and graph plotting.

Requires Hydraulics Bench Service unit F1-10

Look at the video :





Armfield C4-MKII Venturi Flume



Armfield C4-MKII Adjustable Undershot Weir

Armfield C4-MKII Broad Crested Weir

Armfield C4-MKII Crump Weir



Armfield C4-MKII Sharp Crested Weir shown in unvented condition

TECHNICAL SPECIFICATIONS

MODELS AND GAUGES SUPPLIED

Venturi flume
Sharp and broad-crested weirs
Crump weir
Adjustable undershot weir
Two Vernier level gauges (Hook and point gauges)

CHANNEL DIMENSIONS

Width 76mm
Height 250mm
Channel Slope Adjustable between -1% and +3%

FEATURES & BENEFITS

Experimental Content

- Study of open channel flume
- Use of hook and point gauges to measure water level
- Learning how to apply force-momentum and steady-flow energy equations to simple flow situations
- Understanding the relationship between water level above the crest of a weir and flow rate over the weir
- Using hydraulic structures to control level, eg siphon spillways
- Understanding sub and super-critical flow and the underlying characteristics of waves
- Hydraulic jump
- Using hydraulic structures for control of flow, eg sluice gate

- Applying and understanding Manning's formula
- Use of a Pitot static tube to measure flow rate and velocity profiles (using optional C4-61)

DETAILED EXPERIMENTAL CONTENT

- To determine the relationship between upstream head and flowrate for water flowing over a Sharp Crested weir
- To calculate the discharge coefficient and to observe the flow patterns obtained over a Sharp Crested weir
- To determine the relationship between upstream head and flowrate for water flowing over a Broad Crested weir (long base weir)
- To calculate the discharge coefficient and to observe the flow patterns obtained over a Broad Crested weir (long base weir)
- To determine the relationship between upstream head and flowrate for water flowing over a Crump weir
- To determine the modular limit and to observe the flow patterns obtained over a Crump weir
- To determine the relationship between upstream head and flowrate for water flowing under a sluice gate (undershot weir)
- To calculate the discharge coefficient and to observe the flow patterns obtained under a sluice gate (undershot weir)
- To determine the relationship between upstream head and thrust on a sluice gate (undershot weir) for water flowing under the sluice gate
- To determine the relationship between the specific energy and upstream head for water flowing under an undershot weir
- To investigate the characteristics of a standing wave (the hydraulic jump) produced when water flows beneath an undershot weir and to observe the flow patterns obtained
- To determine the relationship between upstream head and flowrate for water flowing through a Venturi flume
- To calculate the discharge coefficient and to observe the flow patterns obtained through a Venturi flume
- To determine the characteristics and observe the flow patterns obtained for water flowing through a culvert. Requires C4-62
- To observe the flow patterns obtained for water flowing around splitters with different profiles. Requires C4-63
- To observe the flow patterns associated with the flow of water over a dam spillway. Requires C4-64
- To determine the relationship between upstream head and flowrate through a siphon spillway in the "blackwater" fully primed condition. Requires C4-65
- To calculate the discharge coefficient and to observe the operation of the siphon as it primes and de-primed. Requires C4-65
- To determine the relationship between upstream head and flowrate through an air regulated siphon. Requires C4-65
- To calculate the discharge coefficient and to observe the operation of the air regulated siphon as it primes and de-primed. Requires C4-65
- To determine the relationship between upstream head and flowrate beneath a radial gate (Tainter Gate) under different operating conditions. Requires C4-66
- Calculate the discharge coefficient in each position of the radial gate. Requires C4-66
- To observe the flow patterns associated with the flow of water over different bed profiles. Requires C4-68
- To determine the effect of a roughened bed on the depth of water at different flowrates and to obtain appropriate coefficients to satisfy the Manning Formula. Requires C4-69
- To determine the effect of different wave regimes on different flow patterns. Requires C4-67

Software

Optional educational software is available (C4-MKII-ABASIC) offering a complete teaching package of coursework.

The student manually enters data in the software, which can then be used for calculations, data processing and graph plotting.

Accessories





Armfield-C4-61 Pitot Tube and Manometer

Armfield C4-62 Culvert

Armfield C4-63_Flow Splitters X3



Armfield C4-64 Free overflow spillways



Armfield C4-65 Air-regulated siphon



Armfield C4-65 Siphon Spillway

Armfield C4-66 Model radial gate

Armfield C4-67 Wave generator



Armfield C4-MKII Multi-Purpose Teaching Flume Wave on Beach

Armfield C4-68 False floor sections





Armfield C4-69 Artificially Roughened Bed

Ordering specification

- A 76mm wide, 250mm high open channel for use with an F1-10 Hydraulics Bench
- Available in 2.5m and 5.0m working section lengths
- Clear acrylic sides to give visibility of the working section
- A jacking system permits the slope of the channel bed to be adjusted between -1% and +3%
- Inlet tank with flow stilling arrangement
- Includes a Venturi, sharp and broad-crested weirs, 2 vernier level gauges, adjustable undershot weir and crump weir
- Wide range of other models available as accessories.
- Optional flow meter
- Comprehensive instruction manual

Requirements

- **Electrical supply:** The F1-10 requires an electrical supply, please refer to the [F1 data sheet](#) for details.
 - **F1-10-A:** 220-240V / 1ph / 50Hz / 10A
 - **F1-10-B:** 110-120V / 1ph / 60Hz / 10A

- **F1-10-G:** 220V / 1ph / 60Hz / 10A
- **F1-10-2-A:** 220-240V / 1ph / 50Hz / 10A
- **F1-10-2-B:** 110-120V / 1ph / 60Hz / 10A
- **F1-10-2-G:** 220V / 1ph / 60Hz / 10A

Shipping Specifications

PACKED AND CRATED SHIPPING SPECIFICATIONS

C4-MKII-2.5M

Volume: 2.8m³

Gross Weight: 330Kg

C4-MKII-5.0M

Volume: 3.0m³

Gross Weight: 380Kg

Dimensions

C4-MKII-2.5M

Length: 2.91m

Width: 0.62m

Height: 1.46m

C4-MKII-5.0M

Length: 5.41m

Width: 0.62m

Height: 1.46m

Order Codes

C4-MKII-2.5M-10

Multi-purpose Flume, 2.5m, with basic accessories

C4-MKII-5.0M-10

Multi-purpose Flume, 5m, with basic accessories

C4-MKII-ABASIC

Educational Software

YOU MAY ALSO LIKE...



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